

TAMIL NADU STATE BOARD - STANDARD 9 SCIENCE (PHYSICS)

CHAPTER 5: WORK, POWER AND ENERGY

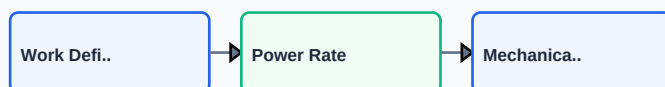
PREMIUM ACADEMIC STUDY SUITE & EXAM GUIDE

Board:	Tamil Nadu State Board (TNSB)
Syllabus:	Samacheer Kalvi Textbook Curriculum aligned
Standard:	Class 9 English Medium
Subject:	Science (Physics)
Chapter Name:	Work, Power and Energy

Learning Objectives:

- Comprehend core theoretical concepts and exact textbook terminology.
- Apply relevant formulas and proofs to solve high-scoring board problems.
- Interpret and sketch vector diagrams or graphical models representing equations.
- Build examination readiness for 1-mark, 2-mark, and 5-mark question sets.

TOPIC ROADMAP: WORK, POWER, AND ENERGY RELATIONS



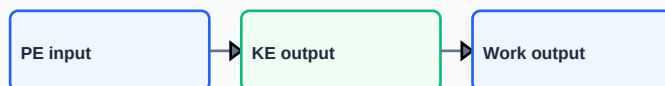
Core Concepts & Definitions

Textbook definitions are essential for scoring fully in the short-answer section of board assessments. Review the primary definitions below:

Term	Official Definition & Context
Work	Work is said to be done by a force when the force produces displacement in the body in the direction of the force. SI unit: Joule (J).
Energy	The capacity to do work, which exists in various forms and can be converted from one form to another. SI unit: Joule (J).
Kinetic Energy	The energy possessed by a body by virtue of its motion, proportional to its mass and the square of its speed.
Potential Energy	The energy possessed by a body by virtue of its position, shape, or configuration relative to a conservative field.
Power	The rate of doing work or rate of energy transfer. SI unit: Watt (W) = 1 Joule/second.

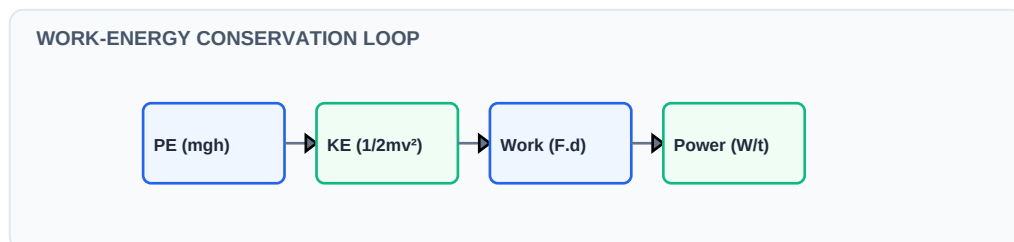
Syllabus Exam Tip: Memorize precise keywords. Examiners award maximum marks when textbook terms and correct symbols are correctly referenced.

GLOSSARY: MECHANICAL ENERGY STAGE CONVERSIONS



Detailed Explanation & Formulas

Here, we analyze the core laws and formulas. Examine the conceptual illustration below to visual the relation between variables:



Essential Formulas, Laws & Identities:

- Work: $W = F \cdot s \cdot \cos \theta$
- Kinetic Energy: $KE = \frac{1}{2} \cdot m \cdot v^2$
- Gravitational Potential Energy: $PE = m \cdot g \cdot h$
- Power: $P = \text{Work} / \text{time} = W / t = F \cdot v$
- Mechanical Energy conservation: Total Energy (E) = KE + PE = constant
- 1 Horsepower (HP) = 746 Watts

Make sure to check the dimensional consistency of all physical quantities and verify that units are correctly stated in final solutions.

Worked Examples

Review these standard board examination-style worked examples to learn proper step layout structures:

Example 1: A porter lifts a suitcase of 15 kg from the ground and puts it on his head 1.5 m above the ground. Calculate the work done.

Step-by-step Solution:

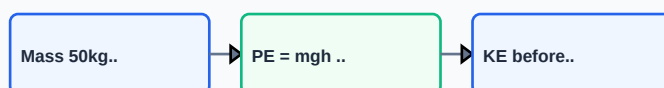
Given: $m = 15 \text{ kg}$, $h = 1.5 \text{ m}$, $g = 9.8 \text{ m/s}^2$. Work done $W = m \cdot g \cdot h = 15 \cdot 9.8 \cdot 1.5 = 220.5 \text{ Joules}$.

Example 2: An electric motor of 2 HP runs for 10 seconds. Calculate the electrical work done.

Step-by-step Solution:

Power in Watts = $2 \cdot 746 = 1492 \text{ W}$. Work done $W = \text{Power} \cdot \text{time} = 1492 \cdot 10 = 14920 \text{ Joules}$.

WORKED: ENERGY CONSERVATION OF FALLING MASS



HOTS (Higher Order Thinking Skills) Challenge

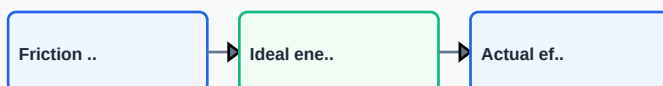
Challenge yourself with these complex, multi-concept problems designed to test deeper conceptual understanding and mathematical reasoning:

HOTS Challenge 1: A heavy box is pushed against a wall, and the pusher becomes exhausted but the box does not move. Has any mechanical work been done? Explain.

Analytical Solution Strategy:

No. Mechanical work requires a non-zero displacement in the direction of the force ($W = F \cdot s$). Since the box remains stationary (displacement $s = 0$), the mechanical work done is zero. The energy expended by the pusher is dissipated internally as physiological heat.

HOTS: MACHINE EFFICIENCY LOSSES AND POWER LIMITS



Practice Worksheet

Test your skills by attempting these sample exam problems independently. Draw diagrams where appropriate:

Q1 (1-Mark): Define work and state the conditions for work to be positive, negative, or zero.

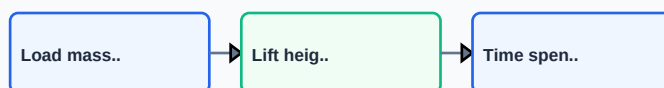
Q2 (2-Mark): State the Work-Energy Theorem.

Q3 (2-Mark): Prove that the potential energy of a body at a height h is mgh .

Q4 (5-Mark): A body of mass 10 kg is dropped from a height of 20 m. Calculate its KE and PE at: (i) start, (ii) 10 m height, (iii) just before hitting the ground.

Q5 (5-Mark): Differentiate between commercial unit of energy (kWh) and Joules.

PRACTICE: CALCULATING POWER OUTPUT WORKSHEET



Real-World Applications & Connections

This chapter connects directly with modern industrial engineering, computation, and natural systems in numerous ways:

Application Case: Hydroelectric Power Plants

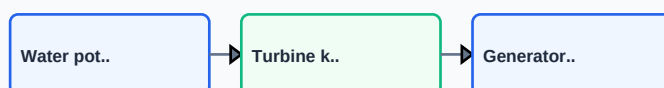
Water stored in dams (Potential Energy) flows down (Kinetic Energy) to rotate turbines and generate electricity.

Application Case: Automobile Engines

Engine performance is rated in Horsepower (HP) to specify maximum torque and load capabilities.

Cross-Curricular Link: Concepts learned in this unit are heavily applied in other subjects like Physics (equations of motion, field vectors) and Data Analytics.

APPLICATION: HYDROELECTRIC POWER STATION CONVERSION FLOW



Revision Summary & Strategy

Use this checklist to self-evaluate your exam preparation status before final tests:

Key Recall Tips:

- ✓ Always write down 'Given Parameters' first to ensure partial marks are locked in.
- ✓ Pay attention to signs (+ / -) when performing algebraic operations or force mappings.
- ✓ Check that you have written the correct units (e.g. Joules, Tesla, cm^3 , N) for the final numeric answer.
- ✓ Draw neat diagrams using a sharp pencil and ruler for geometry or circuit layouts.

Syllabus Checklist:

- ☐ Read and memorized core definitions on Page 2.
- ☐ Reviewed critical formulas and relations on Page 3.
- ☐ Solved worked examples on Page 4.
- ☐ Attempted worksheet problems on Page 6.

CONGRATULATIONS! YOU HAVE COMPLETED THIS SYLLABUS STUDY GUIDE!

Practice consistently, verify answers with standard textbook methods, and take the topic test in the Nexora Learning app to gauge progress.
Good luck!

REVISION: WORK AND ENERGY CONSERVATION LOOP

